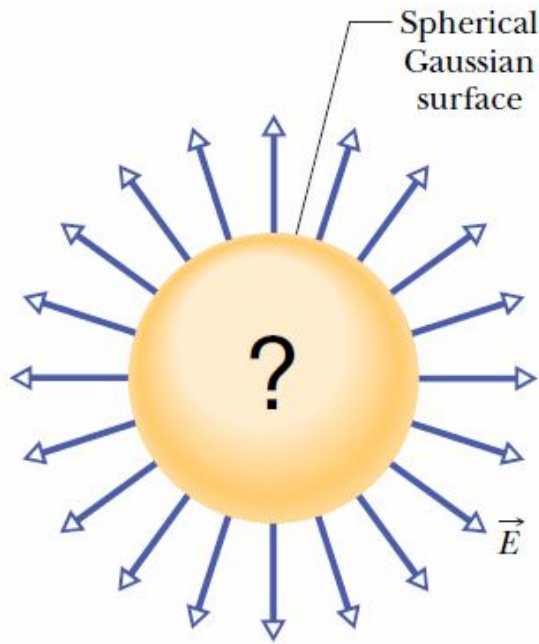


Halliday & Resnick's
Fundamentals of Physics
Extended Edition **Wiley**
Jearl Walker

Gauss's Law
Chapter 23

23.1 What is Physics?:

Gauss' law relates the electric fields at points on a (closed) Gaussian surface to the net charge enclosed by that surface.



Gauss' law considers a hypothetical (imaginary) closed surface enclosing the charge distribution.

This **Gaussian surface**, as it is called, can have any shape, but the shape that minimizes our calculations of the electric field is one that mimics the symmetry of the charge distribution.

Fig. 23-1 A spherical Gaussian surface. If the electric field vectors are of uniform magnitude and point radially outward at all surface points, you can conclude that a net positive distribution of charge must lie within the surface and have spherical symmetry.

23.2: Flux

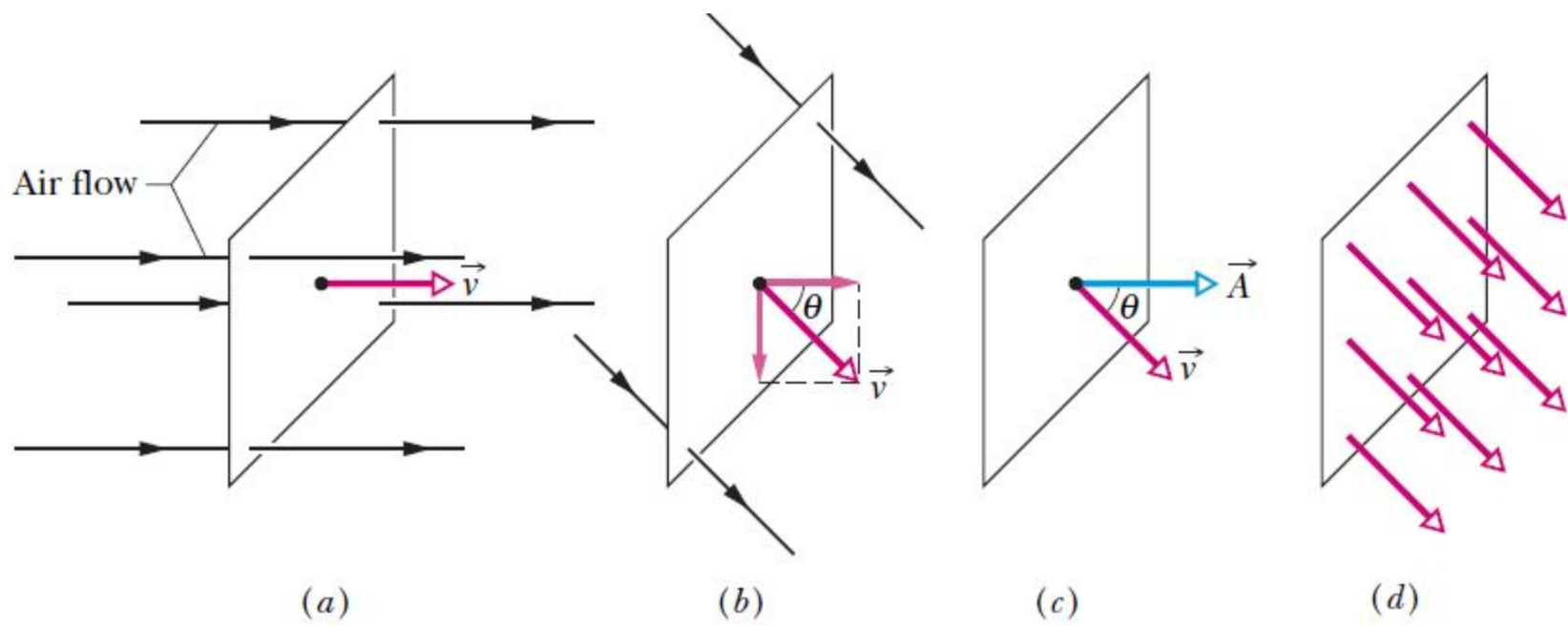


Fig. 23-2 (a) A uniform airstream of velocity is perpendicular to the plane of a square loop of area A . (b) The component of perpendicular to the plane of the loop is $v \cos \theta$, where θ is the angle between $\vec{v}$ and a normal to the plane. (c) The area vector $\vec{A}$ is perpendicular to the plane of the loop and makes an angle θ with $\vec{v}$. (d) The velocity field intercepted by the area of the loop. The rate of volume flow through the loop is $\Phi = (v \cos \theta)A$.

This rate of flow through an area is an example of a flux—a *volume flux* in this situation.

$$\Phi = vA \cos \theta = \vec{v} \cdot \vec{A},$$

23.3: Electric Flux

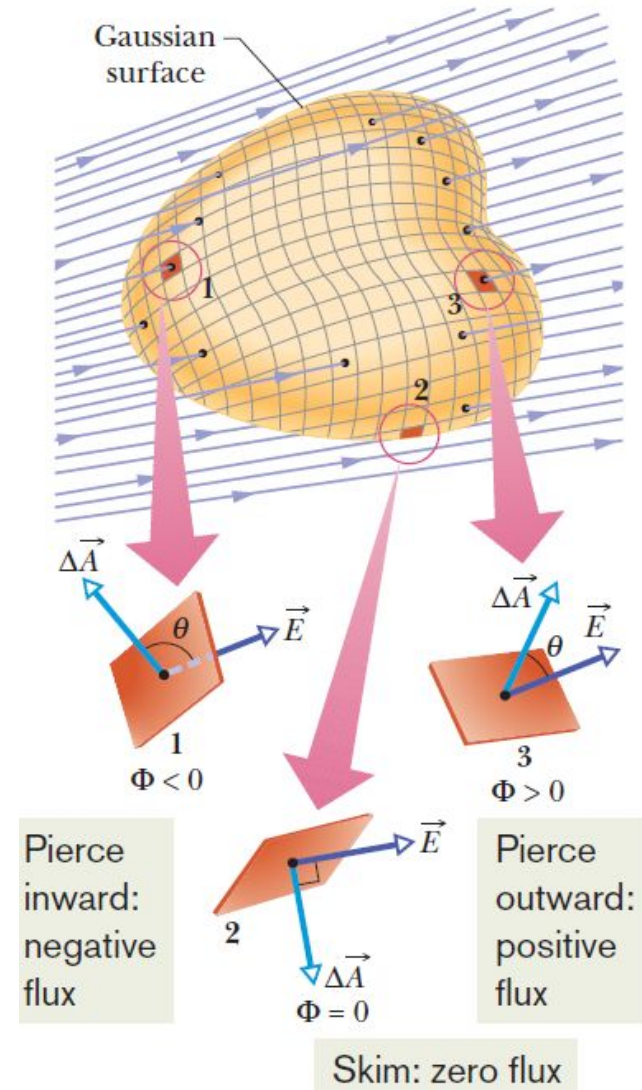
The electric flux through a Gaussian surface is proportional to the net number of electric field lines passing through that surface.

Fig. 23-3 A Gaussian surface of arbitrary shape immersed in an electric field. The surface is divided into small squares of area ΔA . The electric field vectors $\vec{E}$ and the area vectors $\Delta\vec{A}$ for three representative squares, marked 1, 2, and 3, are shown.

$$\Phi = \sum \vec{E} \cdot \Delta\vec{A}.$$

The exact definition of the flux of the electric field through a closed surface is found by allowing the area of the squares shown in Fig. 23-3 to become smaller and smaller, approaching a differential limit dA . The area vectors then approach a differential limit $d\vec{A}$. The sum of Eq. 23-3 then becomes an integral:

$$\Phi = \oint \vec{E} \cdot d\vec{A} \quad (\text{electric flux through a Gaussian surface}).$$



Example, Flux through a closed cylinder, uniform field:

Figure 23-4 shows a Gaussian surface in the form of a cylinder of radius R immersed in a uniform electric field $\vec{E}$, with the cylinder axis parallel to the field. What is the flux Φ of the electric field through this closed surface?

KEY IDEA

We can find the flux Φ through the Gaussian surface by integrating the scalar product $\vec{E} \cdot d\vec{A}$ over that surface.

Calculations: We can do the integration by writing the flux as the sum of three terms: integrals over the left cylinder cap a , the cylindrical surface b , and the right cap c . Thus, from Eq. 23-4,

$$\begin{aligned}\Phi &= \oint \vec{E} \cdot d\vec{A} \\ &= \int_a \vec{E} \cdot d\vec{A} + \int_b \vec{E} \cdot d\vec{A} + \int_c \vec{E} \cdot d\vec{A}. \quad (23-5)\end{aligned}$$

For all points on the left cap, the angle θ between $\vec{E}$ and $d\vec{A}$ is 180° and the magnitude E of the field is uniform. Thus,

$$\int_a \vec{E} \cdot d\vec{A} = \int E(\cos 180^\circ) dA = -E \int dA = -EA,$$

where $\int dA$ gives the cap's area A ($= \pi R^2$). Similarly, for the

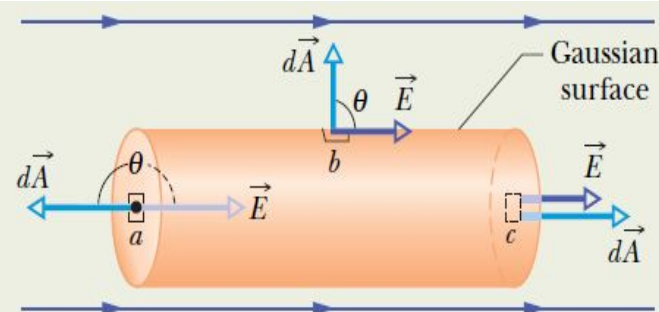


Fig. 23-4 A cylindrical Gaussian surface, closed by end caps, is immersed in a uniform electric field. The cylinder axis is parallel to the field direction.

right cap, where $\theta = 0$ for all points,

$$\int_c \vec{E} \cdot d\vec{A} = \int E(\cos 0) dA = EA.$$

Finally, for the cylindrical surface, where the angle θ is 90° at all points,

$$\int_b \vec{E} \cdot d\vec{A} = \int E(\cos 90^\circ) dA = 0.$$

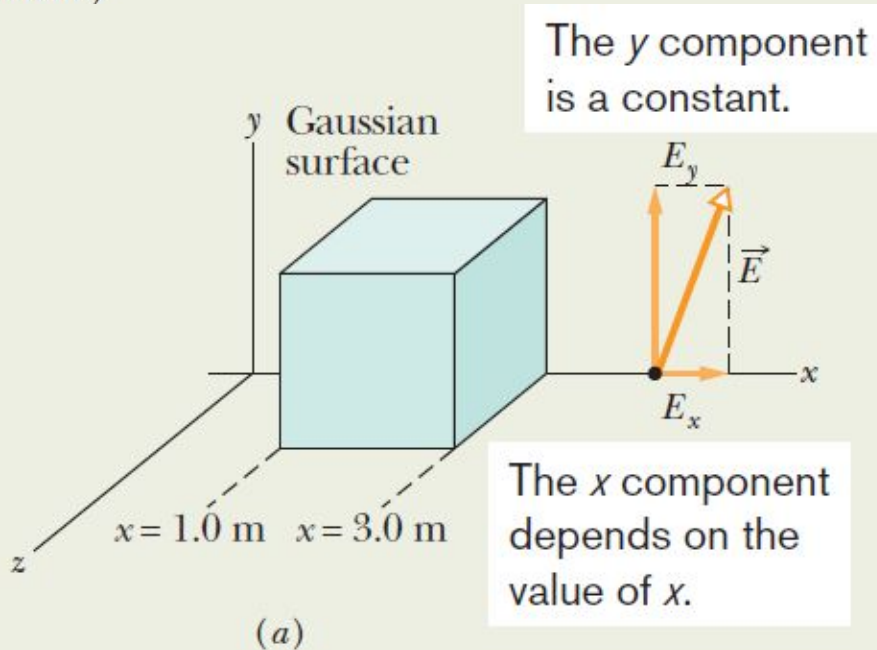
Substituting these results into Eq. 23-5 leads us to

$$\Phi = -EA + 0 + EA = 0. \quad (\text{Answer})$$

The net flux is zero because the field lines that represent the electric field all pass entirely through the Gaussian surface, from the left to the right.

Example, Flux through a closed cube, Non-uniform field:

A *nonuniform* electric field given by $\vec{E} = 3.0x\hat{i} + 4.0\hat{j}$ pierces the Gaussian cube shown in Fig. 23-5a. (E is in newtons per coulomb and x is in meters.) What is the electric flux through the right face, the left face, and the top face? (We consider the other faces in another sample problem.)



Right face: An area vector $\mathbf{A}$ is always perpendicular to its surface and always points away from the interior of a Gaussian surface. Thus, the vector for any area element $d\mathbf{A}$ (small section) on the right face of the cube must point in the positive direction of the x axis. The most convenient way to express the vector is in unit-vector notation,

$$d\vec{A} = dA\hat{i}.$$

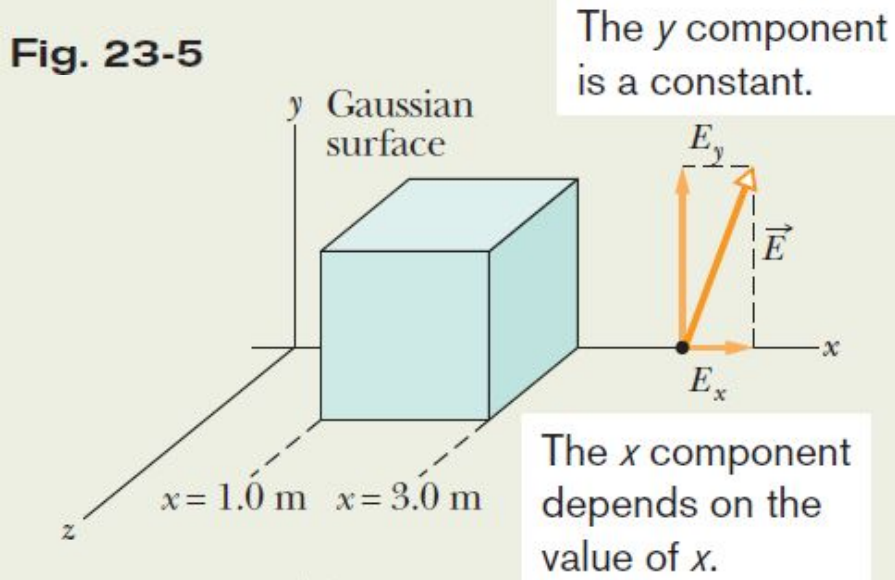
$$\begin{aligned}\Phi_r &= \int \vec{E} \cdot d\vec{A} = \int (3.0x\hat{i} + 4.0\hat{j}) \cdot (dA\hat{i}) \\ &= \int [(3.0x)(dA)\hat{i} \cdot \hat{i} + (4.0)(dA)\hat{j} \cdot \hat{i}] \\ &= \int (3.0x dA + 0) = 3.0 \int x dA.\end{aligned}$$

Although x is certainly a variable as we move left to right across the figure, because the right face is perpendicular to the x axis, every point on the face has the same x coordinate. (The y and z coordinates do not matter in our integral.) Thus, we have

$$\Phi_r = 3.0 \int (3.0) dA = 9.0 \int dA. = (9.0 \text{ N/C})(4.0 \text{ m}^2) = 36 \text{ N} \cdot \text{m}^2/\text{C}. \quad (\text{Answer})$$

Example, Flux through a closed cube, Non-uniform field:

A *nonuniform* electric field given by $\vec{E} = 3.0x\hat{i} + 4.0\hat{j}$ pierces the Gaussian cube shown in Fig. 23-5a. (E is in newtons per coulomb and x is in meters.) What is the electric flux through the right face, the left face, and the top face? (We consider the other faces in another sample problem.)



Left face: The procedure for finding the flux through the left face is the same as that for the right face. However, two factors change. (1) The differential area vector $d\vec{A}$ points in the negative direction of the x axis, and thus $d\vec{A} = -dA\hat{i}$ (Fig. 23-5d). (2) The term x again appears in our integration, and it is again constant over the face being considered. However, on the left face, $x = 1.0$ m. With these two changes, we find that the flux Φ_l through the left face is

$$\Phi_l = -12 \text{ N} \cdot \text{m}^2/\text{C}. \quad (\text{Answer})$$

Top face: The differential area vector $d\vec{A}$ points in the positive direction of the y axis, and thus $d\vec{A} = dA\hat{j}$ (Fig. 23-5e). The flux Φ_t through the top face is then

$$\begin{aligned} \Phi_t &= \int (3.0x\hat{i} + 4.0\hat{j}) \cdot (dA\hat{j}) \\ &= \int [(3.0x)(dA)\hat{i} \cdot \hat{j} + (4.0)(dA)\hat{j} \cdot \hat{j}] \\ &= \int (0 + 4.0 dA) = 4.0 \int dA \\ &= 16 \text{ N} \cdot \text{m}^2/\text{C}. \quad (\text{Answer}) \end{aligned}$$

23.4 Gauss's Law:

Gauss' law relates the net flux of an electric field through a closed surface (a Gaussian surface) to the net charge q_{enc} that is enclosed by that surface.

$$\epsilon_0 \Phi = q_{enc} \quad (\text{Gauss' law}).$$

$$\epsilon_0 \oint \vec{E} \cdot d\vec{A} = q_{enc} \quad (\text{Gauss' law}).$$

The net charge q_{enc} is the algebraic sum of all the enclosed positive and negative charges, and it can be positive, negative, or zero.

If q_{enc} is positive, the net flux is outward; if q_{enc} is negative, the net flux is inward.

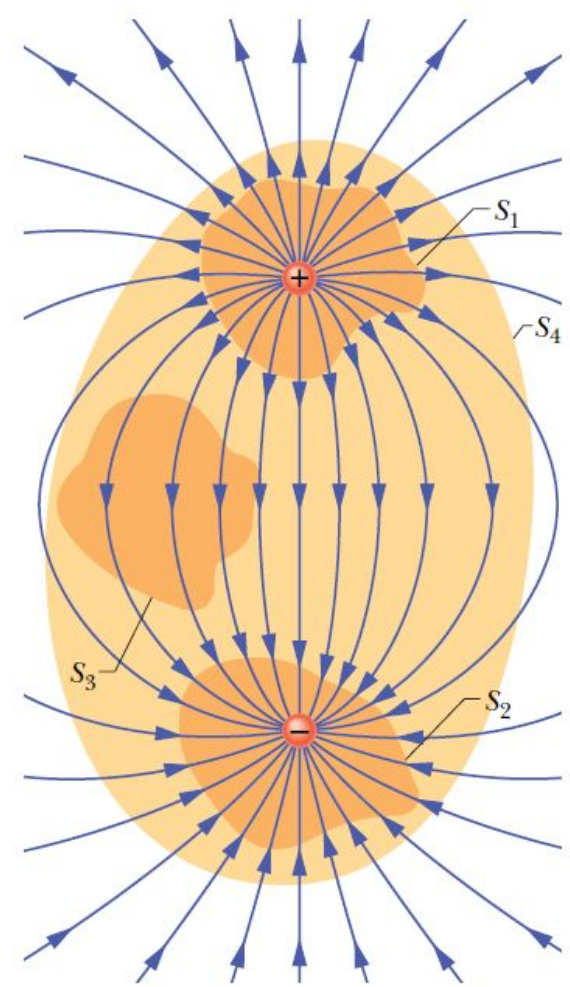


Fig. 23-6 Two point charges, equal in magnitude but opposite in sign, and the field lines that represent their net electric field. Four Gaussian surfaces are shown in cross section. Surface S_1 encloses the positive charge. Surface S_2 encloses the negative charge. Surface S_3 encloses no charge. Surface S_4 encloses both charges and thus no net charge.

Example, Relating the net enclosed charge and the net flux:

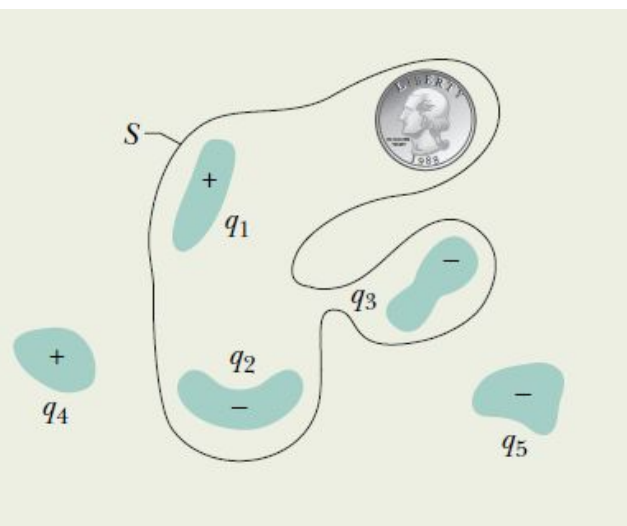


Fig. 23-7 Five plastic objects, each with an electric charge, and a coin, which has no net charge. A Gaussian surface, shown in cross section, encloses three of the plastic objects and the coin.

Figure 23-7 shows five charged lumps of plastic and an electrically neutral coin. The cross section of a Gaussian surface S is indicated. What is the net electric flux through the surface if $q_1 = q_4 = +3.1$ nC, $q_2 = q_5 = -5.9$ nC, and $q_3 = -3.1$ nC?

KEY IDEA

The *net* flux Φ through the surface depends on the *net* charge q_{enc} enclosed by surface S .

Calculation: The coin does not contribute to Φ because it is neutral and thus contains equal amounts of positive and negative charge. We could include those equal amounts, but they would simply sum to be zero when we calculate the *net* charge enclosed by the surface. So, let's not bother. Charges q_4 and q_5 do not contribute because they are outside surface S . They certainly send electric field lines

through the surface, but as much enters as leaves and no net flux is contributed. Thus, q_{enc} is only the sum $q_1 + q_2 + q_3$ and Eq. 23-6 gives us

$$\begin{aligned}\Phi &= \frac{q_{\text{enc}}}{\epsilon_0} = \frac{q_1 + q_2 + q_3}{\epsilon_0} \\ &= \frac{+3.1 \times 10^{-9} \text{ C} - 5.9 \times 10^{-9} \text{ C} - 3.1 \times 10^{-9} \text{ C}}{8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2} \\ &= -670 \text{ N} \cdot \text{m}^2/\text{C}. \quad (\text{Answer})\end{aligned}$$

The minus sign shows that the net flux through the surface is inward and thus that the net charge within the surface is negative.

23.5 Gauss's Law and Coulomb's Law:

Figure 23-8 shows a positive point charge q , around which a concentric spherical Gaussian surface of radius r is drawn. Divide this surface into differential areas dA .

The area vector $d\mathbf{A}$ at any point is perpendicular to the surface and directed outward from the interior.

From the symmetry of the situation, at any point the electric field, $\mathbf{E}$, is also perpendicular to the surface and directed outward from the interior.

Thus, since the angle θ between $\mathbf{E}$ and $d\mathbf{A}$ is zero, we can rewrite Gauss' law as

$$\epsilon_0 \oint \vec{E} \cdot d\vec{A} = \epsilon_0 \oint E dA = q_{\text{enc}}.$$

$$\epsilon_0 E \oint dA = q.$$

$$\epsilon_0 E(4\pi r^2) = q$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}.$$

This is exactly what Coulomb's law yielded.

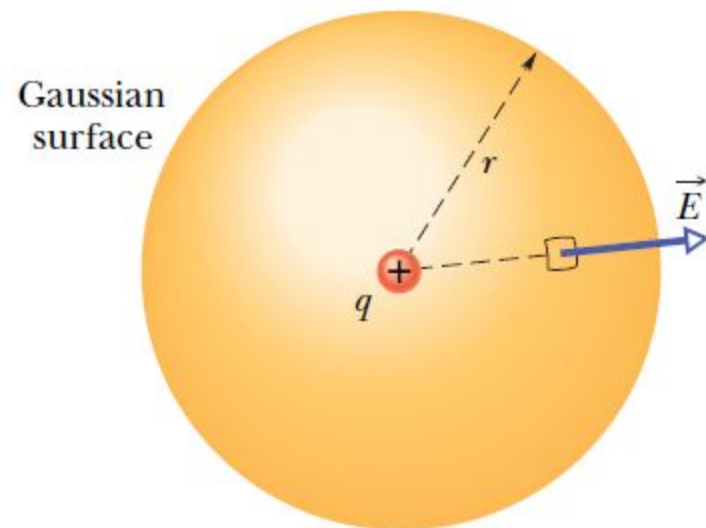


Fig. 23-8 A spherical Gaussian surface centered on a point charge q .

23.6 A Charged Isolated Conductor; The External Electric Field:

The electric field just outside the surface of a conductor is easy to determine using Gauss' law.

Consider a section of the surface that is small enough to neglect any curvature and thus the section is considered flat.

A tiny cylindrical Gaussian surface is embedded in the section as in Fig. 23-10: One end cap is fully inside the conductor, the other is fully outside, and the cylinder is perpendicular to the conductor's surface.

The electric field E at and just outside the conductor's surface must also be perpendicular to that surface.

We assume that the cap area A is small enough that the field magnitude E is constant over the cap.

Then the flux through the cap is EA , and that is the net flux Φ through the Gaussian surface.

The charge q_{enc} enclosed by the Gaussian surface lies on the conductor's surface in an area A . If σ is the charge per unit area, then q_{enc} is equal to σA .

$$\epsilon_0 EA = \sigma A.$$

$$E = \frac{\sigma}{\epsilon_0} \quad (\text{conducting surface}).$$

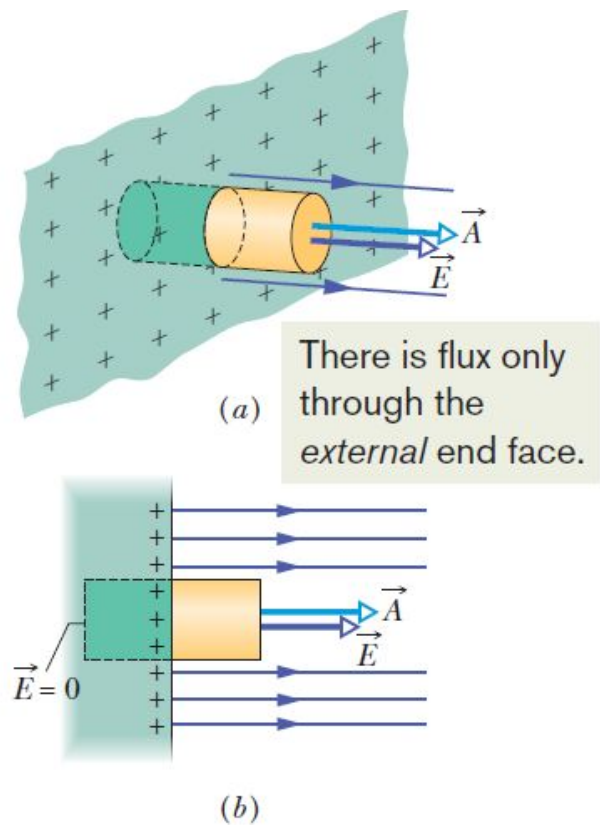


Fig. 23-10 (a) Perspective view and (b) side view of a tiny portion of a large, isolated conductor with excess positive charge on its surface. A (closed) cylindrical Gaussian surface, embedded perpendicularly in the conductor, encloses some of the charge. Electric field lines pierce the external end cap of the cylinder, but not the internal end cap. The external end cap has area A and area vector $\vec{A}$.

23.7 Applying Gauss's Law and Cylindrical Symmetry:

Figure 23-12 shows a section of an infinitely long cylindrical plastic rod with a uniform positive linear charge density λ .

Let us find an expression for the magnitude of the electric field $\vec{E}$ at a distance r from the axis of the rod.

At every point on the cylindrical part of the Gaussian surface, must have the same magnitude E and (for a positively charged rod) must be directed radially outward.

The flux of $\vec{E}$ through this cylindrical surface is

$$\Phi = EA \cos \theta = E(2\pi rh) \cos 0 = E(2\pi rh).$$

$$\epsilon_0 \Phi = q_{\text{enc}},$$

$$\epsilon_0 E(2\pi rh) = \lambda h,$$

$$E = \frac{\lambda}{2\pi\epsilon_0 r} \quad (\text{line of charge}).$$

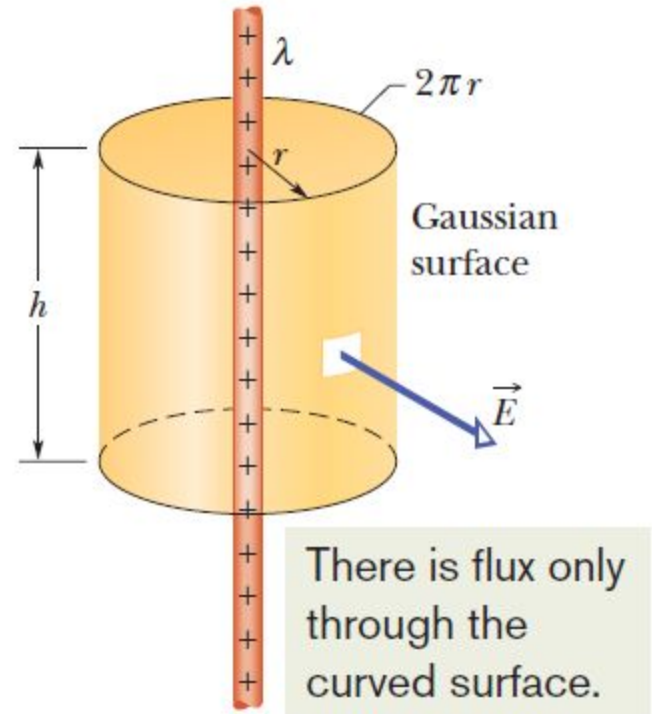


Fig. 23-12 A Gaussian surface in the form of a closed cylinder surrounds a section of a very long, uniformly charged, cylindrical plastic rod.

Example, Gauss's Law and an upward streamer in a lightning storm:

Upward streamer in a lightning storm. The woman in Fig. 23-13 was standing on a lookout platform in the Sequoia National Park when a large storm cloud moved overhead. Some of the conduction electrons in her body were driven into the ground by the cloud's negatively charged base (Fig. 23-14a), leaving her positively charged. You can tell she was highly charged because her hair strands repelled one another and extended away from her along the electric field lines produced by the charge on her.

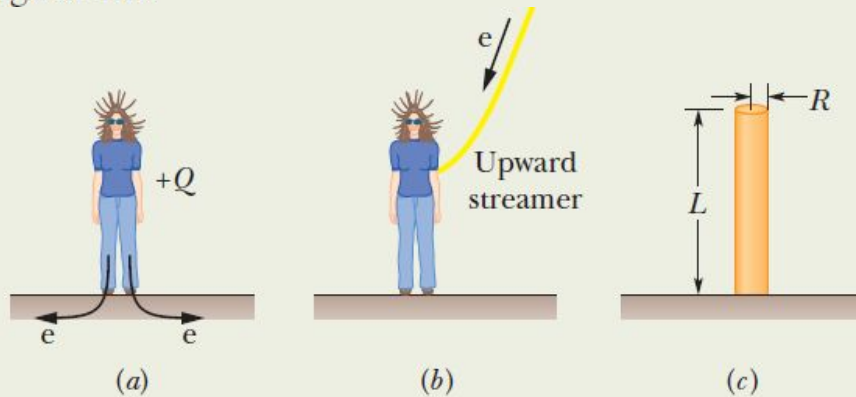


Fig. 23-14 (a) Some of the conduction electrons in the woman's body are driven into the ground, leaving her positively charged. (b) An upward streamer develops if the air undergoes electrical breakdown, which provides a path for electrons freed from molecules in the air to move to the woman. (c) A cylinder represents the woman.

Let's model her body as a narrow vertical cylinder of height $L = 1.8 \text{ m}$ and radius $R = 0.10 \text{ m}$ (Fig. 23-14c). Assume that charge Q was uniformly distributed along the cylinder and that electrical breakdown would have occurred if the electric field magnitude along her body had exceeded the critical value $E_c = 2.4 \text{ MN/C}$. What value of Q would have put the air along her body on the verge of breakdown?

KEY IDEA

Because $R \ll L$, we can approximate the charge distribution as a long line of charge. Further, because we assume that the charge is uniformly distributed along this line, we can approximate the magnitude of the electric field along the side of her body with Eq. 23-12 ($E = \lambda/2\pi\epsilon_0 r$).

Calculations: Substituting the critical value E_c for E , the cylinder radius R for radial distance r , and the ratio Q/L for linear charge density λ , we have

$$E_c = \frac{Q/L}{2\pi\epsilon_0 R},$$

or

$$Q = 2\pi\epsilon_0 R L E_c.$$

Substituting given data then gives us

$$\begin{aligned} Q &= (2\pi)(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(0.10 \text{ m}) \\ &\quad \times (1.8 \text{ m})(2.4 \times 10^6 \text{ N/C}) \\ &= 2.402 \times 10^{-5} \text{ C} \approx 24 \mu\text{C}. \end{aligned} \quad (\text{Answer})$$

23.8 Applying Gauss's Law, Planar Symmetry

Non-conducting Sheet:

Figure 23-15 shows a portion of a thin, infinite, nonconducting sheet with a uniform (positive) surface charge density σ .

A sheet of thin plastic wrap, uniformly charged on one side, can serve as a simple model.

We need to find the electric field a distance r in front of the sheet. A useful Gaussian surface is a closed cylinder with end caps of area A , arranged to pierce the sheet perpendicularly as shown.

From symmetry, $\mathbf{E}$ must be perpendicular to the sheet and hence to the end caps.

Since the charge is positive, $\mathbf{E}$ is directed away from the sheet.

There is no flux through this portion of the Gaussian surface. Thus $\mathbf{E} \cdot d\mathbf{A}$ is simply $E dA$, and

$$\begin{aligned} \epsilon_0 \oint \vec{E} \cdot d\vec{A} &= q_{\text{enc}} \\ \epsilon_0(EA + EA) &= \sigma A. \end{aligned}$$

Here σA is the charge enclosed by the Gaussian surface.

Therefore,

$$E = \frac{\sigma}{2\epsilon_0} \quad (\text{sheet of charge}).$$

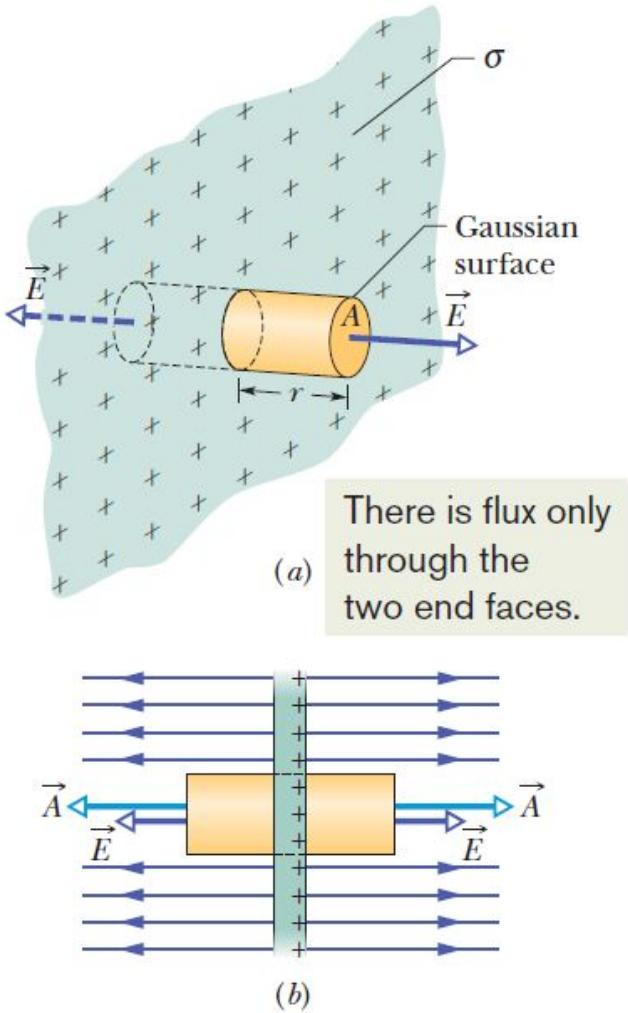


Fig. 23-15 (a) Perspective view and (b) side view of a portion of a very large, thin plastic sheet, uniformly charged on one side to surface charge density σ . A closed cylindrical Gaussian surface passes through the sheet and is perpendicular to it.

23.8 Applying Gauss's Law, Planar Symmetry

Two Conducting Plates:

Figure 23-16a shows a **cross section of a thin, infinite conducting plate with excess positive charge**. The plate is thin and very large, and essentially all the excess charge is on the two large faces of the plate.

If there is no external electric field to force the positive charge into some particular distribution, it will spread out on the two faces with a uniform surface charge density of magnitude σ_1 .

Just outside the plate this charge sets up an electric field of magnitude $E = \sigma_1 / \epsilon_0$

Figure 23-16b shows **an identical plate with excess negative charge** having the same magnitude of surface charge density. Now the electric field is directed toward the plate. $E = \sigma_1 / \epsilon_0$

If we arrange **for the plates of Figs. 23-16 a and b** to be close to each other and parallel (Fig. 23-16c), the excess charge on one plate attracts the excess charge on the other plate, and all the excess charge moves onto the inner faces of the plates as in Fig. 23-16c.

With twice as much charge now on each inner face, the new surface charge density, σ , on each inner face is twice σ_1 .

Thus, the electric field at any point between the plates has the magnitude

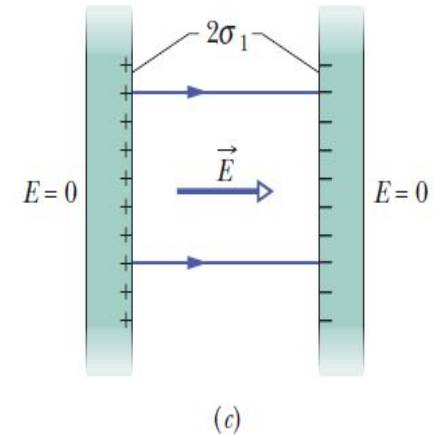
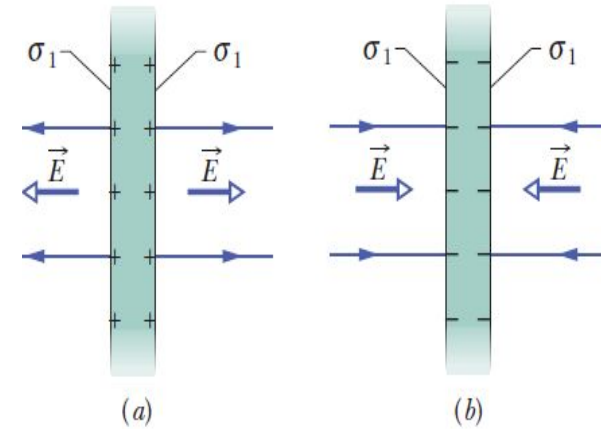


Fig. 23-16 (a) A thin, very large conducting plate with excess positive charge. (b) An identical plate with excess negative charge. (c) The two plates arranged so they are parallel and close.

$$E = \frac{2\sigma_1}{\epsilon_0} = \frac{\sigma}{\epsilon_0}.$$

Example, Electric Field:

Figure 23-17a shows portions of two large, parallel, nonconducting sheets, each with a fixed uniform charge on one side. The magnitudes of the surface charge densities are $\sigma_{(+)} = 6.8 \mu\text{C}/\text{m}^2$ for the positively charged sheet and $\sigma_{(-)} = 4.3 \mu\text{C}/\text{m}^2$ for the negatively charged sheet.

Find the electric field $\vec{E}$ (a) to the left of the sheets, (b) between the sheets, and (c) to the right of the sheets.

KEY IDEA

With the charges fixed in place (they are on nonconductors), we can find the electric field of the sheets in Fig. 23-17a by (1) finding the field of each sheet as if that sheet were isolated and (2) algebraically adding the fields of the isolated sheets via the superposition principle. (We can add the fields algebraically because they are parallel to each other.)

Calculations: At any point, the electric field $\vec{E}_{(+)}$ due to the positive sheet is directed *away* from the sheet and, from Eq. 23-13, has the magnitude

$$E_{(+)} = \frac{\sigma_{(+)}}{2\epsilon_0} = \frac{6.8 \times 10^{-6} \text{ C/m}^2}{(2)(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)} = 3.84 \times 10^5 \text{ N/C}.$$

Similarly, at any point, the electric field $\vec{E}_{(-)}$ due to the negative sheet is directed *toward* that sheet and has the magnitude

$$E_{(-)} = \frac{\sigma_{(-)}}{2\epsilon_0} = \frac{4.3 \times 10^{-6} \text{ C/m}^2}{(2)(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)} = 2.43 \times 10^5 \text{ N/C}.$$

Figure 23-17b shows the fields set up by the sheets to the left of the sheets (*L*), between them (*B*), and to their right (*R*).

The resultant fields in these three regions follow from the superposition principle. To the left, the field magnitude is

$$\begin{aligned} E_L &= E_{(+)} - E_{(-)} \\ &= 3.84 \times 10^5 \text{ N/C} - 2.43 \times 10^5 \text{ N/C} \\ &= 1.4 \times 10^5 \text{ N/C}. \end{aligned} \quad (\text{Answer})$$

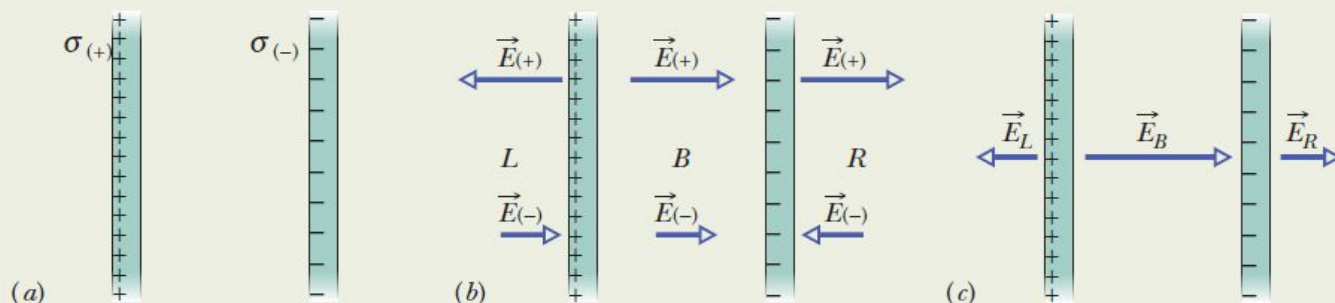
Because $E_{(+)}$ is larger than $E_{(-)}$, the net electric field $\vec{E}_L$ in this region is directed to the left, as Fig. 23-17c shows. To the right of the sheets, the electric field has the same magnitude but is directed to the right, as Fig. 23-17c shows.

Between the sheets, the two fields add and we have

$$\begin{aligned} E_B &= E_{(+)} + E_{(-)} \\ &= 3.84 \times 10^5 \text{ N/C} + 2.43 \times 10^5 \text{ N/C} \\ &= 6.3 \times 10^5 \text{ N/C}. \end{aligned} \quad (\text{Answer})$$

The electric field $\vec{E}_B$ is directed to the right.

Fig. 23-17 (a) Two large, parallel sheets, uniformly charged on one side. (b) The individual electric fields resulting from the two charged sheets. (c) The net field due to both charged sheets, found by superposition.



23.9 Applying Gauss's Law, Spherical Symmetry:

- A shell of uniform charge attracts or repels a charged particle that is outside the shell as if all the shell's charge were concentrated at the center of the shell.
- If a charged particle is located inside a shell of uniform charge, there is no electrostatic force on the particle from the shell.

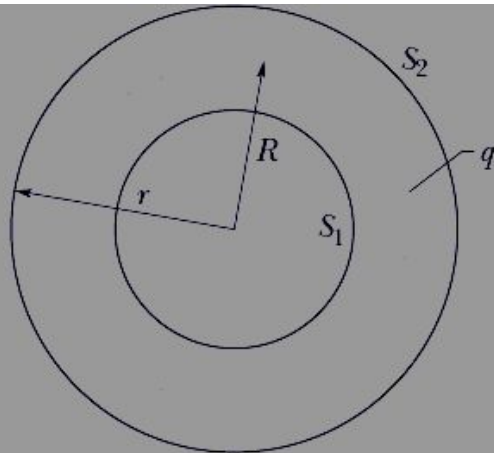


Fig. 23-18 A thin, uniformly charged, spherical shell with total charge q , in cross section. Two Gaussian surfaces S_1 and S_2 are also shown in cross section. Surface S_2 encloses the shell, and S_1 encloses only the empty interior of the shell.

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \quad (\text{spherical shell, field at } r \geq R).$$

$$E = 0 \quad (\text{spherical shell, field at } r < R),$$

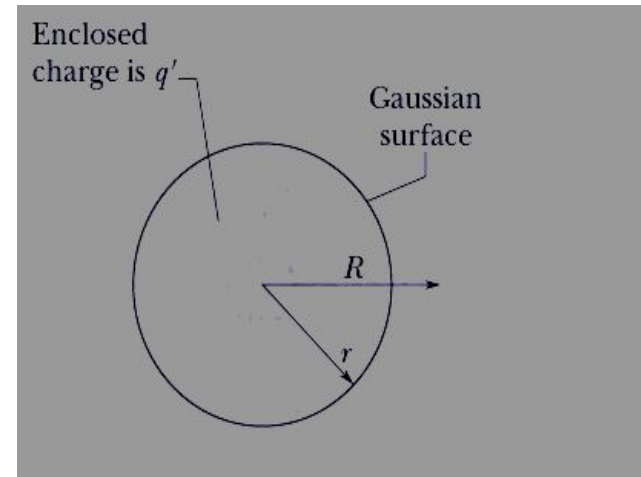


Fig. 23-19 The dots represent a spherically symmetric distribution of charge of radius R , whose volume charge density ρ is a function only of distance from the center. The charged object is not a conductor, and therefore the charge is assumed to be fixed in position. A concentric spherical Gaussian surface with $r > R$ is shown.

$$E = \frac{1}{4\pi\epsilon_0} \frac{q'}{r^2} \quad (\text{spherical distribution, field at } r \leq R).$$

$$E = \left(\frac{q}{4\pi\epsilon_0 R^3} \right) r \quad (\text{uniform charge, field at } r \leq R).$$